

Optimizing ADC Linkers via LSTM Reinforcement Learning

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Abstract

Antibody-drug conjugates (ADCs) are targeted cancer therapies, made up of a monoclonal antibody (mAb), cytotoxic payload, and linker. To address the need for rapid and optimized linker generation, we developed a long short-term memory (LSTM) model which implements reinforcement learning to generate chemically valid and structurally optimal ADC linkers for specific antibody-conjugate pairs. The model was initially trained on a dataset of approximately 13,000 real and synthetic ADC linkers. Sequences were generated as self-referencing embedding strings (SELFIES) which guarantees chemically valid molecules, as opposed to SMILES which rely on complex syntax that is difficult for generative models to learn and apply. Linker specific motifs including heads, cleavable regions and tail groups were given their own unique tokens to ensure fidelity and to assess whether linkers were structurally complete. The generative model was then trained via reinforcement learning (RL) to optimize linker generation for desirable chemical properties and ADC specific conditions such as antibody and payload attachment and cancer target. The RL trained model produces chemically valid and unique linkers above 90% for all generated sequences and achieves a low Tanimoto similarity of less than 20%. Average chemical properties related to ease of synthesis and solubility are comparable to transformer based linker generation models. Additionally, compared to pre-RL training, the post-RL model generally produced better results for all chemical properties while maintaining high structural uniqueness. This work provides a foundational model for generating novel ADC linkers which could expedite ADC development for modern precision medicine. This model still lags behind transformer based approaches in generating structures with high drug-likeness (QED) and future work will aim to address these shortcomings. Long-term objectives for this model are to adopt a graph neural network (GNNs) architecture, utilizing transformer based predictions, and to extend desirable linker properties to include metabolite toxicity in ADC-specific tissue environments. Software package provided at: [Repo Link](#)

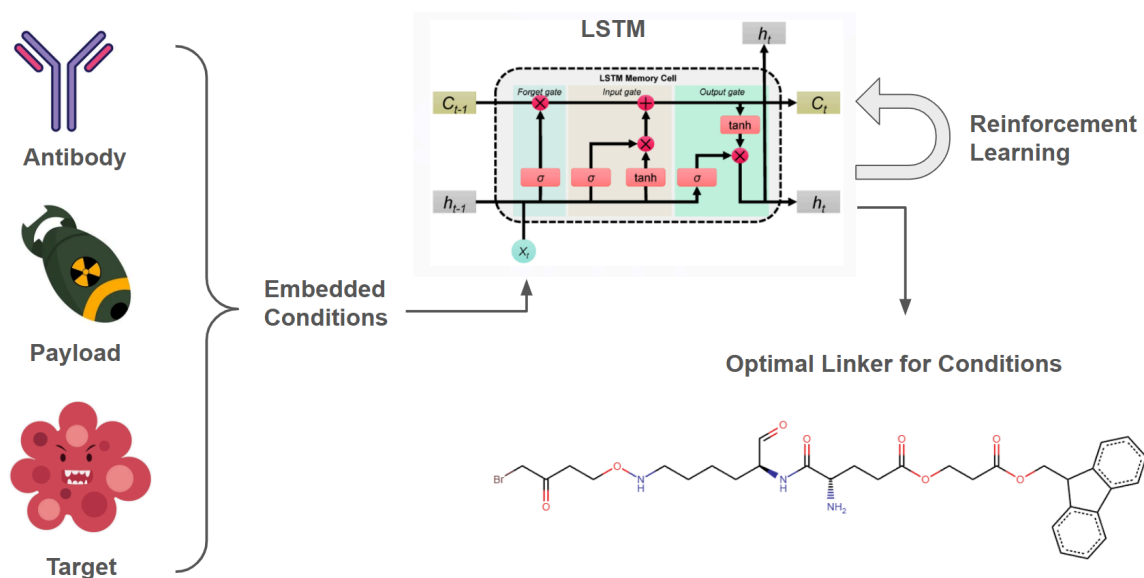


Figure 1. Graphical Abstract of LSTM-RL Model. ADC relevant conditions are embedded to each token generated to produce relevant structures for those conditions. Reinforcement learning applies linker-specific rewards related to chemical properties, ADC conditional adherence and presence of linker motifs to ultimately produce more optimal linker structures.

Introduction

Antibody-drug conjugates (ADCs) represent a novel class of targeted therapies, each consisting of a monoclonal antibody (mAb), cytotoxic payload, and linker. The mAb enables the ADC to have high specificity for cancer cell surface antigens, allowing precise delivery of the cytotoxic payload and avoiding off-target toxicity.¹ This improves upon traditional chemotherapy, which cannot distinguish between cancerous and healthy tissue, greatly compromising on quality of life for the patient. Due to the specificity of the ADC mechanism, the function of the linker between mAb and payload is crucial in the success of the treatment. The linker must keep the mAb and payload connected while traveling to the target, then depending on the therapeutic application must be able to release the payload via some cleavage mechanism once the mAb is bound to the appropriate target cell². Typical linker development involves slow, combinatorial lab-based approaches that are costly to perform. Additionally, research groups must be extremely confident in their ADC structure due to high cost and risk associated with clinical trials. With the aid of Artificial Neural Networks (ANNs) we are able to rapidly generate optimal linkers for common mAb-drug pairs, to target specific cancers. This advancement decreases the time used in laboratory screening, and provides valuable insights that will lead to clinical success. This LSTM model was trained on aggregate data from the Database of Antibody-drug Conjugates³ and Linker-GPT model dataset, and then through RL was able to generate optimal linker candidates. This work will help to establish a machine learning, data-driven approach for guiding linker design and accelerate the success of next-generation ADC therapies.

Implementation

Data Collection. The training dataset was constructed from the Linker-GPT⁴ linker data and the ADCdb³ clinical ADC database. The Linker-GPT dataset contained standard chemical properties like synthetic accessibility (SA), drug-likeness (QED) and solubility (LogP). These structures were cross-referenced against ADCdb to include antibody, payload and cancer target information for each linker. The combined dataset consists of approximately 3,500 unique linker structures. Additional chemical property features were generated for the model's reward system including transmembrane permeability (TPSA) and fraction of SP3 carbon (CSP3), which is another proxy for drug-likeness. Ultimately, this dataset was too small to adequately train the LSTM, so additional synthetic linkers were generated via breaking of synthetically interesting chemical substructures (BRICS)¹¹. The real linker data was fragmented via BRICS and linker specific motifs were identified. These motifs include reactive head groups, cleavable spacers and protected tails. These motifs were randomly combined to generate an additional 10,000 linkers for the training dataset.

Exploratory Data Analysis. The training data was assessed to determine separability of different linker types based on the chemical property features. This analysis revealed a core challenge in that the majority of linkers are chemically very similar. On average, they have a similar size, a similar hydrophobicity, a similar membrane permeability, etc. Additionally, these aggregate features did little to differentiate what was unique about linker structures as it pertained to specific antibodies, payloads and cancers. The exploratory data analysis ultimately showed that aggregate feature-based separation is weak. This further necessitated our need for a generative model that could learn what made these linkers structurally unique, in order to optimize for ADC specific conditions and ideal chemical properties.

own predictions can lead to overreliance on the teacher. Scheduled learning combines a mix of forced teaching and independent predictions to address this shortcoming and typically produces a more robust model prior to starting RL training.

Several rounds of pre-RL training were performed to fine tune the model's predictions. Initially the model was trained on the huge synthetic dataset to learn general molecular grammar. Then the model was trained on subsequent datasets that were more narrow in scope. This included the combined real and synthetic data, just the real data, and finally a filtered version of the real data containing the best linker candidates. In general, at each stage the learning rate was reduced as the scope of the task became more narrow. This process helped to establish a strong baseline understanding for SELFIES and then fine-tune the model's understanding of ADC specific chemistry.

Reinforcement Learning. Reinforcement learning is an area of machine learning in which a model learns and updates its policy for action based on delayed reward signals to guide its behavior toward an ultimate goal. In our case, the ultimate goal was to generate SELFIES strings which were not only syntactically valid, but optimized for properties required for ADCs. Reinforcement learning relies on calculating an advantage for each training episode using the Proximal Policy Optimization (PPO) algorithm¹⁰. For each token generated by the LSTM, a secondary critic model predicts how that token will impact the final reward. This critic model is a simple multi-layer perceptron neural network designed just to make these reward estimations. During initial training, an end token was appended to the end of each sequence so that the model could learn when to stop a sequence. During RL, when an end token is encountered a reward is calculated.

There were three categories of rewards. The first were ADC structural rewards that looked for the presence of ADC specific motif tokens. Sequences that contained all three required motifs were rewarded more heavily than those that contained fewer. Secondly, chemical property scores were calculated. These were based on SA, QED, LogP, etc. An aggregate of these scores was passed back to the model which worked to maximize or minimize various properties. An important innovation in calculating chemical property rewards centered around chemical deprotection. Many of these structures contain bulky tail protecting groups that are required during synthesis but are later removed when the linker is incorporated into the ADC. It was necessary to calculate SA with these present, but calculate the remaining drug-likeness and solubility properties with these motifs removed. This was done through mapping of protecting groups to their equivalent deprotected state. Lastly, conditional rewards were calculated. During token generation, conditional embeddings for antibody, payload and cancer target are appended to each token. This is meant to guide generation towards structures specific to these conditions. In order to judge how well structures adhered to these conditions, another multi-layer perceptron was utilized. This model was just trained on ADC conditions and produces a prediction based on how much it thinks a structure will fall into a certain conditional class. The certainty of the prediction was the reward for the model. Fine-tuning the balance between these three different reward types was the single most important hyperparameter to tune for RL training. Subtle changes to this balance would teach the model to cheat and just focus on one objective because its reward outweighed the others. Ultimately this is still an area of focus in further refining the models performance.

Logit Biasing. One of the main issues with using ML to generate molecules is that in organic chemistry, nearly everything is just carbon, oxygen and nitrogen. The model quickly learns that the safest

prediction is to just insert another carbon or oxygen. In addition to this being boring, it ignores the primary objective of generating actual linkers which require specific motifs to be present. To address this, logit biasing was implemented to directly manipulate the logits of motif tokens before making a prediction¹². At each step of the generation, our model recorded whether each motif type had already appeared. Based on this status and current length of the sequence, a bias strength was applied to the logits of specific motif tokens. For example, if we had already generated a head motif but no cleavable motif, the logits corresponding to the cleavable tokens would be increased making it more likely to select them. This method of biasing guides the sampling process by modifying the probability of selecting different motifs while preserving stochasticity. This allows us to maintain structural diversity in our generated sequences. Ultimately, this approach enables our model to reliably generate structurally valid ADC linkers, while maintaining the flexibility to explore novel structural variations.

Results

Generating Linkers. The LSTM model allows the user to generate linkers based on one of four antibody, payload and cancer target combinations. The training data only consisted of linkers that had these specified combinations which is why we were forced to adhere to them for linker generation. In future work, addressing this limitation within the dataset would provide a huge improvement to the flexibility of the model. An example output from the Cetuximab, Perumycin, Triple Negative Breast Cancer conditional combination has been provided (Fig. 3). During generation, 100 linker candidates are produced and the top scoring linkers for SA and QED are returned to the user in a report. Visually, these linkers are all structurally unique and contain the required head, cleavable and tail motifs needed for ADC linkers. The novelty and chemical properties of generated linkers are discussed in the subsequent section.

Antibody: Cetuximab
Payload: Puromycin
Cancer: Triple negative breast cancer

Linker	SA	QED	TPSA	LogP	CSP3
A	2.97	0.42	100.57	2.21	0.18
B	3.54	0.25	99.21	2.90	0.17
C	2.98	0.16	81.70	3.35	0.52
D	3.31	0.19	125.42	4.38	0.44

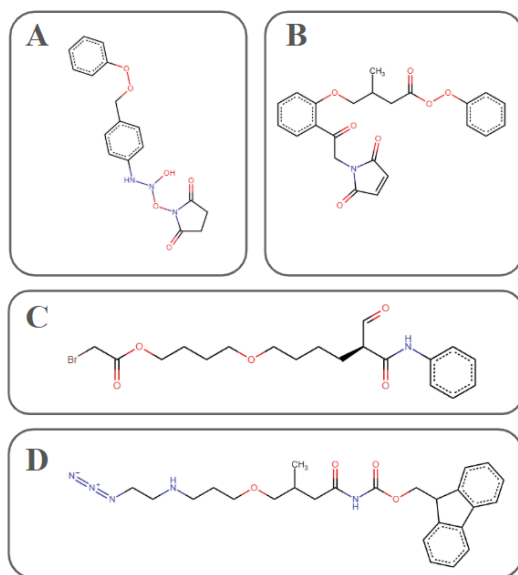


Figure 3. LSTM Linker Candidate Generation. 100 linkers were generated with specific antibody, payload and cancer conditions and the top candidates based on chemical properties were reported.

Model Performance. Overall, our model was able to successfully generate valid, novel, and unique linkers, and reinforcement learning improved average SA, QED, LogP, TPSA, and CSP3 scores relative to

the pre-RL model. The model was able to achieve 92.40% structural validity, 100% novelty, and 95.70% uniqueness in the linkers that were produced post-reinforcement learning (Table 1). A structurally valid molecule was successfully converted from SELFIES back to SMILES. Novelty indicates the rate at which generated linkers appear in the original training set. Uniqueness refers to the duplicate rate for generated linkers amongst themselves. For each of these metrics, 1000 linkers were generated in alignment with the performance strategy in the Linker-GPT report. These scores are higher than the validity, novelty, and uniqueness scores from the Linker-GPT model, which are 89.40%, 99.7%, and 81.40% respectively. The Tanimoto similarity score is also low, being at 26.87% pre-reinforcement learning, and 19.61% post-reinforcement learning. A low Tanimoto Similarity score indicates that many of our linkers are structurally distinct from each other. Two linkers might be unique if they vary by a single carbon spacer, but in reality these are functionally identical. The Tanimoto score accounts for this by comparing the morgan fingerprints amongst structures. This further confirms that our model is generating truly unique, novel linker structures.

Metric	Pre-RL	Post-RL	Linker-GPT
Validity	94.43%	92.40%	89.40%
Novelty	100.00%	100.00%	99.7%
Uniqueness	97.60%	95.70%	81.40%
Tanimoto Similarity	26.87%	19.61%	-
SA	4.05 $\pm$ 0.90	4.09 $\pm$ 0.76	2.65 - 3.10
QED	0.12 $\pm$ 0.11	0.22 $\pm$ 0.12	0.41 - 0.61
LogP	3.66 $\pm$ 2.02	1.01 $\pm$ 1.40	< 5.0
TPSA	140.94 $\pm$ 76.55	90.16 $\pm$ 34.75	-
CSP3	0.23 $\pm$ 0.19	0.59 $\pm$ 0.21	-

Table 1. Results from pre- and post-reinforcement learning, and Linker-GPT reward metrics.

The LSTM model also performed quite well regarding chemical properties of linkers (Table 1). A relatively low SA score of 4.09 $\pm$ 0.76 was achieved post-RL, although being slightly worse than Linker-GPT's SA score range of 2.65 - 3.10. However below 5.0 for SA is still reasonable for ease of synthesis. Additionally, LogP was below the desired threshold defined in the Linker-GPT paper. The reward for this property was based on minimizing the score as much as possible, but it appears the model may have gone too far. A LogP of around 3.0 is typically preferred and we are currently well below that at 1.01 $\pm$ 1.40¹³. In the future, a reward that focuses on maintaining an appropriate range might be more effective. QED was one metric where the model fell short relative to the Linker-GPT model. Although our QED improved slightly pre and post-RL training, the score is lower than our target benchmark of 0.4. This metric is an aggregate score of several chemical properties related to drug-likeness. In the future, it would make sense to deconstruct this aggregate into individual rewards to keep track of where the model is failing specifically and adjust rewards and hyperparameters accordingly to address those shortcomings.

The additional metrics that we included to assess performance, including membrane permeability and fraction SP3, also saw improvements from RL training. Membrane permeability is critical for ADCs that require the drug to enter the cancer cell prior to cleavage of the payload. Typically 140 is the threshold for good permeability and we are comfortably below this with our generated linkers at 90.16 ± 34.75 . Additionally, CSP3 is another metric that is often used for drug-likeness. Seeing this metric trend in a positive direction following RL indicates that our linkers have become more drug-like. This also highlights the advantage of deconstructing a metric like QED into its individual parts will be effective when designing future chemical property rewards. As a whole, the model performed quite well in generating linkers with favorable chemical properties.

Conditional Adherence. The last objective for this model was to not only produce linkers with desirable chemical properties, but to also adhere to conditions specific to their application. These conditions pertained to the antibody, payload and cancer target for the ADC. It was observed during training that our average reward improved modestly over time, as did the Kullback-Leibler (KL) divergence from the base model (Fig. 4). This means that the model learned to produce better linkers and is distinct from the starting model. To confirm that this divergence from the base model retained knowledge of the conditional embeddings we visualized the predictions via UMAP. This approach visualized how distinct the generated sequences were based upon their ADC conditions by projecting the latent hidden state vectors of the LSTM onto two dimensional space. Initially, to assess how well the model had retained its understanding of the conditional embedding following RL training, the training linker data was passed through the model and the latent vector was extracted. The outputs show distinct clustering for each of our four possible ADC condition combinations (Fig. 5). This confirms that the model retained its understanding of these conditions following RL.

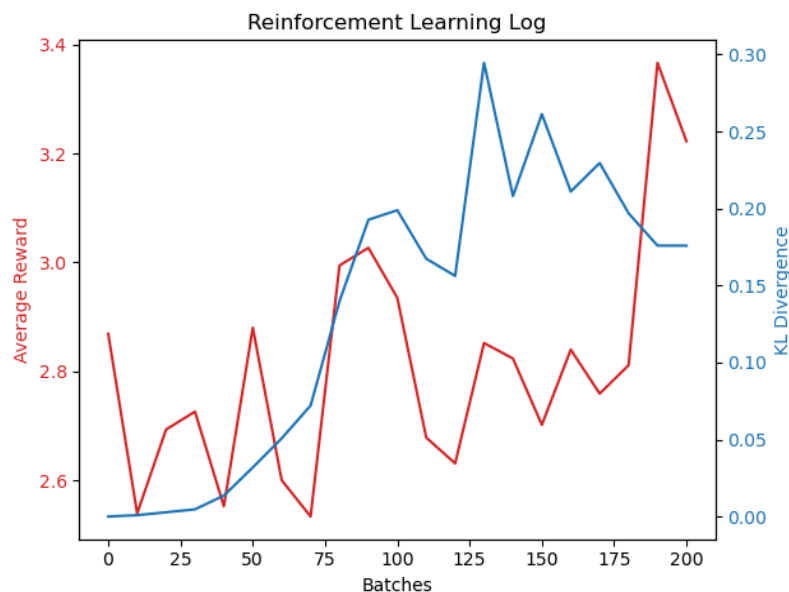


Figure 4. Average Reward and KL Divergence during RL. Rewards and KL divergence were recorded during RL for over 200 sequence batches consisting of 1,000,000 tokens generated.

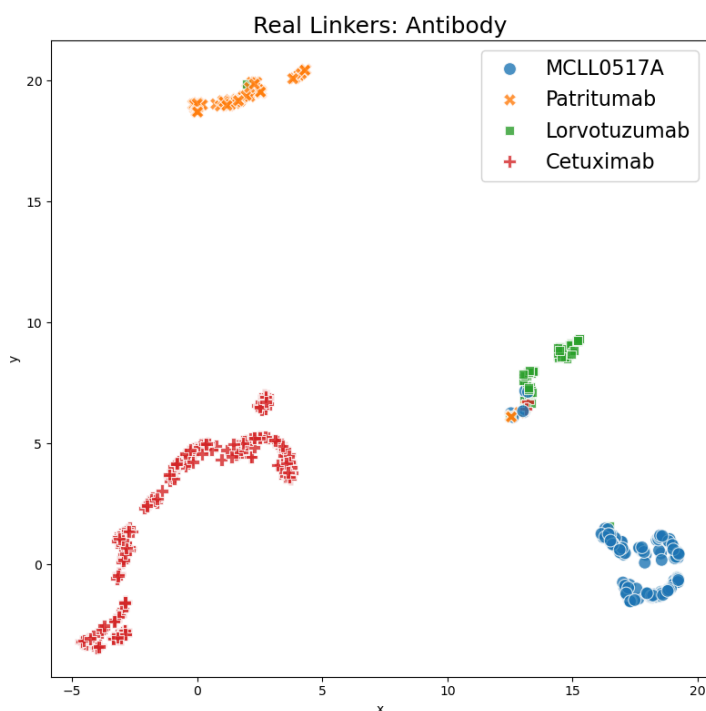


Figure 5. Post-RL LSTM Hidden State from Real Linkers. The training data was fed through the RL trained LSTM model and the latent vector of the model's hidden state was projected via UMAP. The result shows that understanding of conditional embeddings was retained after RL training.

UMAP visualizations were then produced for generated sequences (Fig. 6). The objective in this case was not just to confirm that the model remembered the conditions, but to confirm that the generated linkers were distinct for those conditions. In plot 6.A, we see that the separation between classes is far less distinct compared to the original training data. However we do still see some trends with Lorvotuzumab antibody linkers on the left side of the distribution and Patritumab antibodies on the right. Furthermore, comparing just two antibody condition combos shows much better separation between the condition classes (6.B). In general it seems that the model has determined that there are approximately two distinct conditional classes with a fair amount of overlap in the middle. Lorvotuzumab and MCLL05117A produce similar linkers, as do Patritumab and Cetuximab produce similar linkers. The model has done surprisingly well at retaining its understanding of these conditions and these embeddings seem to provide some influence over the linkers generated. That being said, the linker generation appears to be mostly focused on achieving good chemical properties, not on optimizing for these conditions. Thankfully, because we already have a method by which we can reward based on ADC conditional likeness, this issue can most likely be addressed by further tuning the model's reward system during RL training.

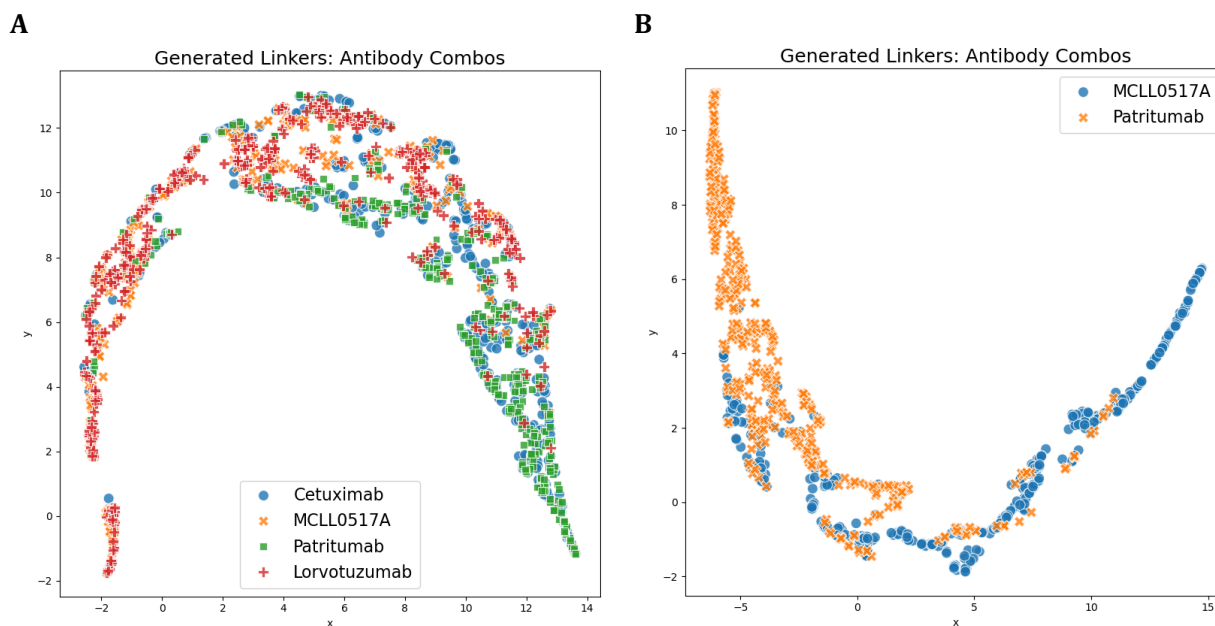


Figure 5. UMAP visualization of generated linkers for antibody combinations. A) Generated linkers for four antibody conditions: cetuximab, MCLL0517A, patritumab, lorvotuzumab. B) Generated linkers for two antibody conditions: MCLL0517A and patritumab.

Discussion

Overall, the model has performed quite well at optimizing important chemical properties and generating novel linkers influenced by ADC specific conditions. In contrast with more advanced neural network architectures like transformer based models that have been used for linker generation, our performance is comparable and we have expanded our implementation to include conditional driven generation. That being said, there are several areas of the current implementation that still need to be improved.

Currently, QED is much lower than what is typically required for new drug candidates. While this single metric does not fully encompass all the important properties for ADC linkers, it is still used as a benchmark for drug candidates and could be the difference between a linker candidate getting selected for clinical trials. As previously mentioned, one way that we could improve this metric is by deconstructing the aggregate score into its individual parts. By doing this we could more effectively monitor which components are over or underperforming during training. This additional granularity would make it easier to effectively tune rewards to optimize for QED.

Another limitation is in regard to the dataset used for training. The limited size of the training data necessitated the use of synthetic linker data to provide enough data for the LSTM model to adequately train. Obviously it would have been far more ideal to only train on real linker data. Looking into other ADC data sources and aggregating those into our current dataset would likely provide a solid boost to the models performance and produce a better baseline model before starting RL training. The second limitation of the dataset was that it only contained four specific antibody, payload and cancer target

combinations for all the real linkers. This is not ideal because it severely limits the applications for which our linkers can be used. If linker data for other conditions is not available there are several other approaches that could make the model more flexible. By assessing the properties of these different conditions regarding the antibody protein structure or the payload structure, we could assign a similarity score to new ADC combinations that the model was never trained on. If the new antibody is similar to one that the model has already seen then we could apply the known embedding to get a good approximation. This still has its shortcomings and assumes that there are only four unique ADC groups which is likely an oversimplification. Ultimately, the main bottleneck to unlocking better performance and flexibility is in the training data available.

One final limitation and area of improvement is in the model architecture itself. For universally all sequence generation tasks, like in LLMs, transformer models are used. Transformers utilize self-attention which allows the model to assess the relationship of all previously generated tokens simultaneously for each prediction. In contrast, an LSTM relies on remembering what happened before, not actively reassessing all relationships. For long sequences with complex relationships like we have with linker generation, we are pushing the limits of the LSTM architecture. Additionally, string representations of molecules, while easy to parse into tokens, do not fully capture what a molecule is. Graph neural networks (GNNs) have been used extensively in machine learning related to molecular structures because they retain spatial information. While out of the scope of what we could accomplish in the time frame for this project, our ideal model would be a generative transformer-GNN architecture that better captures relationships and important three dimensional properties.

In conclusion, the LSTM-RL model described within this report is able to successfully produce novel, chemically optimized linkers for specific ADC applications. Developing better ADC therapies is critical for improving outcomes for cancer patients. The cost of R&D and uncertainty in clinical trials presents a serious financial risk for pharmaceutical companies. Developing smarter tools to help mitigate some of this risk in drug development is therefore hugely beneficial. Optimizing linker structures for ADCs is just one piece of the puzzle in developing next generation cancer therapies and our goal will be to expand upon this work to improve the current model and to adapt this strategy to antibody and payload development.

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